Project Requirement Document

Full Stack Technology Blog Website using MERN Stack

1. Project Description

The project involves developing a technology blog website where the company can regularly publish articles related to technology, software development, emerging trends, industry updates, tutorials and company news.

The website should have a simple content-focused layout that makes it easy for visitors to browse articles, explore different topics and read individual posts. The overall design and content arrangement should take inspiration from Zoho Blog.

2. Website Requirements

Homepage

The homepage should display the latest and featured articles in an organised layout. It should include:

- Header with company logo and navigation menu.
- Featured or highlighted articles.
- Latest blog posts with thumbnail, title, short description and published date.
- Category-based article navigation.
- Search option to find articles.
- Popular or trending articles section.
- Footer with company details, useful links and social media links.

Blog Listing Page

A separate listing page should display all published articles. Visitors should be able to browse articles by category, search using keywords and navigate through the available posts using pagination or a Load More option.

Each article listing should contain a thumbnail image, title, short description, author and published date.

Important Notice:
The use of AI tools to generate or write project code is strictly prohibited. Any candidate found using AI-generated code will be subject to immediate disqualification from the hiring process.

Blog Detail Page

Each article should have its own dedicated page with the following:

- Article title and featured image.
- Author name and published date.
- Estimated reading time.
- Complete article content with support for headings, paragraphs, images, videos, links, tables and code snippets.
- Table of contents for longer articles.
- Social media sharing options.
- Related articles.
- Previous and next article navigation.

3. Content Management System (Admin Panel)

An admin panel should be provided to manage all blog content. The internal team should be able to handle publishing without depending on the development team.

The admin panel should support:

- Creating, editing and deleting blog posts.
- Saving articles as drafts and publishing them.
- Scheduling articles for future publication.
- Uploading featured images and images within articles.
- Creating and managing categories and tags.
- Managing authors.
- Adding SEO title, meta description and URL slug.
- Selecting featured articles.
- Managing published and unpublished posts.

4. Initial Blog Categories

The following categories can be configured initially. The admin should also be able to add or modify categories whenever required.

- Software Development
- Artificial Intelligence & Machine Learning
- Web and Mobile App Development
- Technology & Innovation
- Cybersecurity
- Cloud Computing
- Industry Insights

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5. Design & Technical Requirements

The website should follow a clean corporate design similar to the reference website, with proper typography, spacing and an organised article layout.

It must be responsive across desktop, tablet and mobile devices. Page loading speed, image optimisation and overall reading experience should be considered during development.

The website should also include SEO-friendly URLs, XML sitemap, robots.txt, Open Graph metadata, blog article schema markup and Google Analytics / Search Console integration.

Admin access must be secured with authentication and appropriate user permissions.

6. Project Deliverables

- Responsive technology blog website.
- Homepage, blog listing and individual article pages.
- Category and search functionality.
- Admin panel with complete blog management features.
- SEO configuration.

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